

For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory

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Controlled Release Tablets of
Theophylline
UNICONTIN™ 400/600
CONTINUS™ controlled release system

Description

UNICONTIN™ 400/600 CONTINUS™ tablets for oral administration contain 400/600mg of theophylline BP in a controlled release system which allows a 24-hour dosing interval for appropriate patients.

Unicontin 400:

Each white, round, flat, bevelled edges, scored uncoated tablets engraved with 'MM' logo on one of the facets and 'U/400' on the other.

Unicontin 600:

Each white, biconvex, scored uncoated caplets engraved with 'MM' logo on one of the facets and 'U/600' on the other.

Clinical Pharmacology

Theophylline has two distinct actions: smooth muscle relaxation (e.g. bronchodilation) and suppression of the response of the airways to stimuli (i.e. non-bronchodilator prophylactic effects). Studies suggest that bronchodilation is mediated by the inhibition of two isozymes of phosphodiesterase (PDE III and to a lesser extent, PDE IV) while nonbronchodilator prophylactic actions are probably mediated through one or more different molecular mechanisms, that do not involve inhibition of PDE III or antagonism of adenosine receptors. Theophylline increases the force of contraction of diaphragmatic muscles. This action appears to be due to enhancement of calcium uptake through an adenosine-mediated channel.

Serum Concentration-Effect Relationship: Theophylline is often thought of as a drug which has a particularly well-defined therapeutic range with concentration from 5-20 mcg/ml giving an optimum compromise between efficacy and toxicity. Maintaining the peak serum theophylline concentration between 10-15 mcg/ml, will achieve most of the drug therapeutic benefit with minimal risk & adverse effect. At serum theophylline concentrations >20mcg/mL, both the frequency and severity of adverse reactions increase.

The pharmacokinetics of theophylline vary widely among similar patients and cannot be predicted by age, sex, body weight or other demographic characteristics. In addition, certain concurrent illnesses and alterations in normal physiology and co-administration of other drugs can significantly alter the pharmacokinetic characteristics of theophylline. Within-subject variability in metabolism has also been reported in some studies.

It is, therefore, recommended that serum theophylline concentration be measured frequently in severely ill patients (at 24 hour intervals) and periodically (e.g. at 6-12 month intervals) in patients receiving long-term therapy.

Indications

For the prevention and treatment of bronchial smooth muscle constriction in patients with asthma and chronic bronchitis.

Dosage and Administration

UNICONTIN™ 400/600 mg CONTINUS™ tablets may be taken once a day in the morning or evening. It is recommended that UNICONTIN™ CONTINUS™ be taken with meals. The dose of theophylline must be individualized. Patients should be advised that if they choose to take UNICONTIN™ CONTINUS™ with food it should be taken consistently with food and if they take it in a fasted conditions, it should routinely be taken fasted. It is important that the product when-ever dosed be dosed consistently with or without food.

UNICONTIN™ CONTINUS™ tablets must be swallowed and NOT chewed. UNICONTIN™ 600mg CONTINUS™ tablets may be split. Infrequently, patients receiving UNICONTIN™ CONTINUS™ tablets may pass an intact matrix tablet in the stool or via colostomy. These matrix tablets usually contain little or no residual theophylline. Safety and effectiveness in children under 12 years of age have not been established with UNICONTIN™ CONTINUS™ tablets.

Dosing initiation and titration (as anhydrous theophylline)

A. Patients Without Risk Factors For impaired Clearance:

Titration Step		Children <45 kg (12-15 years)	Children >45 kg and Adults (16-60years)
1.	Starting dosage	12-14mg/kg/day up to a maximum of 300mg/day administered OD	300-400mg/day administered OD
2.	After 3 days, if tolerated, increase dose to :	16mg/kg/day up to a maximum of 400mg/day administered OD	400-600mg/day administered OD
3.	After 3 more days, if tolerated and if needed, Increase dose to :	20mg/kg/day up to a maximum of 600mg/day administered OD	As with all theophylline products doses > 600 mg should be titrated according to blood level.

B. Patients With Risk Factors For impaired Clearance, The Elderly (>60 Years), And Those In Whom it is Not Feasible To Monitor Serum Theophylline Concentrations:

In children 12-15 years of age, the theophylline dose should not exceed 16mg/kg/day up to a maximum of 400mg daily in the presence of risk factors for reduced theophylline clearance or if it is not feasible to monitor serum theophylline concentrations.

In adolescents ≥16 years and adults, including the elderly, the theophylline dose should not exceed 400mg/day in the presence of risk factors for reduced theophylline clearance or if it is not feasible to monitor serum theophylline concentrations.

Dosage adjustment based upon serum theophylline concentration

Peak serum Concentration	Dosage Adjustment
<9.9mcg/mL	If symptoms are <u>not</u> controlled and current dosage is tolerated, increase dose about 25%. Recheck serum concentration after three days for further dosage adjustment.
10-14.9mcg/mL	If symptoms are controlled and current dosage is tolerated, maintain dose and recheck serum concentration at 6-12 month intervals.*
15-19.9mcg/mL	Consider 10% decrease in dose to provide greater margin of safety even if current dosage is tolerated.*
20-24.9mcg/mL	Decrease dose by 25% even if no adverse effects are present. Recheck serum concentration after 3 days to guide further dosage adjustment.

25-30mcg/mL	Skip next dose and decrease subsequent doses at least 25% even if no adverse effects are present. Recheck serum concentration after 3 days to guide further dosage adjustment. If symptomatic, consider whether overdose treatment is indicated.
>30mcg/mL	Treat overdose as indicated. If theophylline is subsequently resumed, decrease dose by at least 50% and recheck serum concentration after 3 days to guide further dosage adjustment.

* Dose reduction and/or serum theophylline concentration measurement is indicated whenever adverse effects are present, physiologic abnormalities that can reduce theophylline clearance occur or a drug that interacts with theophylline is added or discontinued.

Maintenance Therapy

Careful clinical titration is important to assure patient acceptance and safety of the medication. Patients, when stabilised as established by serum theophylline concentration or respiratory function, usually remain controlled without further dosage adjustment. It should be borne in mind however that for reasons stated in the Warnings & Precautions section, dosage adjustments may be necessary. Serum theophylline levels should be measured periodically (at 6 to 12 month intervals) even in clinically controlled patients.

The elderly as well as patients with congestive heart failure, corpulmonale and/or liver disease may have unusually low dosage requirements and thus may experience toxicity even at the recommended dosage.

Do not maintain any dose that is not tolerated.

Contraindications

Patients with a history of hypersensitivity to theophylline or other components in the product; porphyria; concomitant administration with ephedrine in children less than 6 years of age (or less than 22 kg); Theophylline is contraindicated in children under 6 months of age.

Warnings & Precautions

The patient's response to therapy should be carefully monitored – worsening of asthma symptoms requires medical attention.

Due to potential decreased theophylline clearance, dose reduction and monitoring of serum theophylline concentrations may be required in elderly patients and patients with:

- cardiac disease
- hepatic disease
- exacerbations of lung disease
- hypothyroidism
- fever
- viral infections.

Due to potential increased theophylline clearance, dose increase and monitoring of serum theophylline concentrations may be required in patients with hyperthyroidism (and when starting acute hyperthyroidism treatment) and cystic fibrosis.

Theophylline may:

- act as a gastrointestinal tract irritant and increase gastric secretion, therefore caution should be exercised in patients with peptic ulcers;
- exacerbate cardiac arrhythmias and therefore caution should be exercised in patients with cardiac disorders;

- exacerbate frequency and duration of seizures and therefore caution should be exercised in patients with history of seizures and alternative treatment considered.

Use with caution in patients with severe hypertension, or chronic alcoholism.

Caution should be exercised in elderly males with pre-existing partial urinary tract obstruction, such as prostatic enlargement, due to risk of urinary retention.

Particular care is advised in patients suffering from severe asthma who require acute theophylline administration. It is recommended that serum theophylline concentrations are monitored in such situations.

In case of insufficient effect of the recommended dose and in case of adverse events, theophylline plasma concentration should be monitored.

Drug-Drug interactions:

The following increase clearance of theophylline and it may therefore be necessary to increase dosage to ensure a therapeutic effect: aminoglutethimide, carbamazepine, isoprenaline, phenytoin, rifampicin, ritonavir, sulphapyrazole, barbiturates and hypericum perforatum (St John's Wort).

Smoking and alcohol consumption can also increase clearance of theophylline.

The following reduce clearance and a reduced dosage may therefore be necessary to avoid side-effects: aciclovir, allopurinol, carbimazole, cimetidine, clarithromycin, diltiazem, disulfiram, erythromycin, fluconazole, interferon, isoniazid, methotrexate, mexiletine, nizatidine, pentoxifylline, propafenone, propranolol, thiabendazole, verapamil and oral contraceptives (see section "Overdose").

Theophylline has been shown to interact with some quinolone antibiotics including ciprofloxacin and enoxacin which may result in elevated plasma theophylline levels.

The concomitant use of theophylline and fluvoxamine should usually be avoided. Where this is not possible, patients should have their theophylline dose reduced and plasma theophylline should be monitored closely.

Factors such as viral infections, liver disease and heart failure also reduce theophylline clearance (see section "Overdose").

There are conflicting reports concerning the potentiation of theophylline by influenza vaccine and physicians should be aware that interaction may occur resulting in increased serum theophylline levels.

A reduction of dosage may also be necessary in elderly patients.

Thyroid disease or associated treatment may alter theophylline plasma levels.

Concurrent administration of theophylline may:

- inhibit the effect of adenosine receptor agonists (adenosine, regadenoson, dipyridamol) and may reduce their toxicity when used for cardiac perfusion scanning;
- oppose the sedatory effect of benzodiazepines;
- result in the occurrence of arrhythmias with halothane;

- result in thrombocytopenia with lomustine;
- increase urinary lithium clearance.

Therefore these drugs should be used with caution.

Theophylline may decrease steady state phenytoin levels.

Hypokalaemia resulting from beta2 agonist therapy, steroids, diuretics and hypoxia may be potentiated by xanthines. Particular care is advised in patients suffering from severe asthma who require hospitalisation. It is recommended that serum potassium concentrations are monitored in such situations.

Care should be taken in its concomitant use with β -adrenergic agonists, glucagon and other xanthine drugs, as these will potentiate the effects of theophylline.

Co-administration with ketamine may cause reduced convulsive threshold; with doxapram may cause increased CNS stimulation.

The incidence of toxic effects may be enhanced by the concomitant use of ephedrine.

Pregnancy and Lactation:

Pregnancy

Category C- There are no adequate and well controlled studies of the use of theophylline in pregnant women. Theophylline should not be administered during pregnancy unless considered essential by the physician.

Lactation

Theophylline is secreted in breast milk and may be associated with irritability in the infant, therefore it should only be given to breastfeeding women when the anticipated benefits outweigh the risk to the child.

Side Effects

The following adverse drug reactions have been reported in the post-marketing setting for theophylline

Immune system disorders	Anaphylactic reaction
	Anaphylactoid reaction
	Hypersensitivity
Metabolism and nutrition disorders	Hyperuricaemia
Psychiatric disorders	Agitation
	Anxiety
	Insomnia
	Sleep disorder
Nervous system disorders	Convulsions
	Dizziness
	Headache
	Tremor
Cardiac disorders	Atrial tachycardia
	Palpitations
	Sinus tachycardia
Gastrointestinal disorders	Abdominal pain
	Diarrhoea

	Gastric irritation
	Gastro-oesophageal reflux
	Nausea
	Vomiting
Skin and subcutaneous tissue disorders	Pruritus
	Rash
Renal and urinary disorders	Diuresis
	Urinary retention*

* Please refer to section "Warnings & Precautions" as theophylline may induce urinary retention in elderly males with pre-existing partial urinary tract obstruction.

Overdose

Theophylline has a low therapeutic index. Theophylline toxicity is most likely to occur when serum concentrations exceed 20 micrograms/ml and becomes progressively more severe at higher serum concentrations.

Over 3 g could be serious in an adult (40 mg/kg in a child). The fatal dose may be as little as 4.5 g in an adult (60 mg/kg in a child), but is generally higher.

Symptoms

Warning: Serious symptoms may develop as long as 12 hours after overdosage with prolonged release formulations.

Alimentary symptoms: Nausea, vomiting (which is often severe), epigastric pain and haematemesis. Consider pancreatitis if abdominal pain persists.

Neurological symptoms: Restlessness, hypertonia, exaggerated limb reflexes, convulsions, seizures. Coma may develop in very severe cases.

Cardiovascular symptoms: Hypotension. Sinus tachycardia is common. Ectopic beats and supraventricular and ventricular tachycardia may follow.

Metabolic symptoms: Hypokalaemia due to shift of potassium from plasma into cells is common, can develop rapidly and may be severe. Hyperglycaemia, hypomagnesaemia and metabolic acidosis may also occur. Rhabdomyolysis may also occur

Treatment of overdose

Activated charcoal or gastric lavage should be considered if a significant overdose has been ingested within 1-2 hours. Repeated doses of activated charcoal given by mouth can enhance theophylline elimination. Measure the plasma potassium concentration urgently, repeat frequently and correct hypokalaemia. BEWARE! If large amounts of potassium have been given, serious hyperkalaemia may develop during recovery. If plasma potassium is low, then the plasma magnesium concentration should be measured as soon as possible.

In the treatment of ventricular arrhythmias, proconvulsant antiarrhythmic agents such as lignocaine (lidocaine) should be avoided because of the risk of causing or exacerbating seizures.

Measure the plasma theophylline concentration regularly when severe poisoning is suspected, until concentrations are falling. Vomiting should be treated with an antiemetic such as metoclopramide or ondansetron.

Tachycardia with an adequate cardiac output is best left untreated. Beta-blockers may be given in extreme cases but not if the patient is asthmatic.

Control isolated convulsions with intravenous diazepam. Exclude hypokalaemia as a cause.

Particularly in the setting of theophylline overdose induced convulsions, efficacy of some anticonvulsant drugs, such as benzodiazepines, may be reduced through suspected pharmacodynamic interactions.

Effects on ability to drive and use machines

UNICONTINTM CONTINUSTM tablets have no or negligible influence on the ability to drive and use machines.

PHARMACEUTICAL PARTICULARS

Incompatibilities : None Reported

Shelf Life : 36 months

Special Precautions for Storage

Store at or below 30°C, in a dry place, protected from light. Keep out of reach of children.

Packaging

UNICONTINTM 400 : Box of 100 tablets (10 x 10's blister strips).

UNICONTINTM 600 : Box of 100 tablets (10 x 10's blister strips).

Manufactured by:

MODI-MUNDIPHARMA PVT. LTD.

Mfd. at : Modipuram -250 110, UP., India Office :1400, Modi Tower,
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UNICONTINTM 400 : MAL20132342AZ

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TM : Trade Mark

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